

History and SETI

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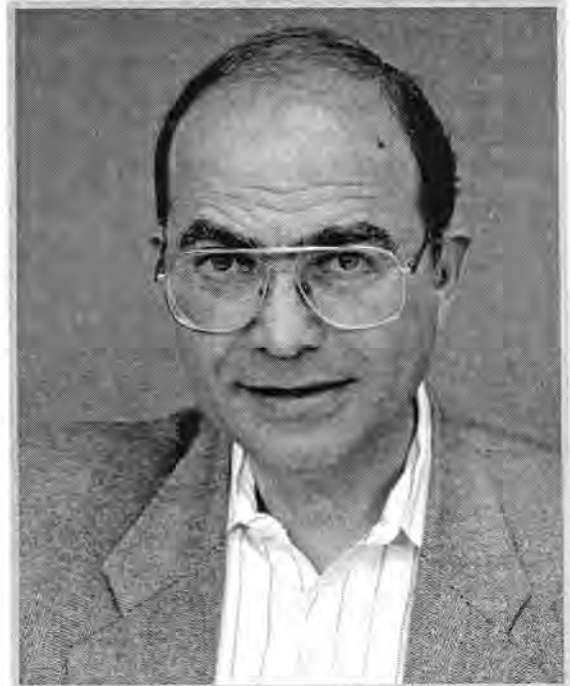
Introduction

NASA's High Resolution Microwave Survey (HRMS) of the stars began on Columbus Day, 1992. The choice of the 500th anniversary of Columbus' landing in the New World as the start of an accelerated effort to detect a signal from an extraterrestrial civilization had immense symbolic significance. Contemporary observers were encouraged to envision an analogy between the European discovery of America and the detection of a signal from beyond the Earth. The analogy has some force: there are evident parallels between Columbus and the architects of the NASA HRMS project in their spirit of adventure, their willingness to accept intellectual risk, and their confidence in the face of the unknown. Columbus shrunk the Earth; he undervalued all pertinent geographical distances to make the gap between Europe and Japan appear traversable by small wooden ships. In a similar manner, the NASA explorers boldly assume proximity of electromagnetically active civilizations, a prospect that would give the search begun last October a chance of success. No doubt like Columbus, the HRMS explorers will find worlds they do not expect. We may call these points the positive aspects of the analogy.

At first glance, the analogy also has some negative aspects. Columbus did not intend to sail to a new world, but to an old emporium. He never fully realized that he had not done what

he had set out to do. Also, irrespective of his misplacing himself by 7000 miles, he came into immediate personal contact with those he called "Indians." That contact allowed direct exchange of culture, information, disease, and artifacts.

Even if ETI signals are found, this aspect of the analogy is baseless. Human listeners will know whence the signals emanate, but we will certainly not meet the senders in person. Indeed, such is the long travel time of even an electromagnetic signal, that its senders might



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have perished long ago. It also seems unlikely that the first detected signal would contain complex information that is immediately meaningful to us, and it is almost certain that we will not be able to exchange information in a two-way dialog with the senders for a very long time.

Discussions of SETI invoke many other analogies, some of which we list and assess. They are invoked because so much of SETI is conjectural. Where lack of knowledge forces conjecture, analogy is a useful (and perhaps the only) guide. However, the proposers and consumers of analogies must be especially careful. Sometimes the negative aspects of an analogy render it useless or seriously misleading. Even the best are imperfect. In particular, analogies drawn from history should be considered as prompts for thinking about the future, not as reliable guides to its course. Nothing in human history is fully analogous to the type of encounter to which SETI research may lead us — a discovery of another technological species.

We begin our assessment of possible human responses to such an encounter with a look at previous comparable incidents from history. The section that follows presents an account of the brief history of SETI research within NASA. Following that, the next section sets that account in a wider social and political context. We conclude with the sorts of analogies commonly used in discussions about SETI and some recommendations for further studies.

A Brief Account of SETI in NASA

From Project Ozma To Project Cyclops

The idea of extraterrestrial life has an extraordinarily long and intriguing history (Dick, 1982; Crowe, 1986; Guthke, 1990). However, only with the dawn of the Space Age did a direct search for life beyond the Earth become scientifically feasible. During the early 1960's, and for the first time in a visual search previously dominated by tantalizing telescopic clues, spacecraft provided a means to seek life *in situ* on Earth-like planets such as Mars.

NASA embraced the direct search for life in the solar system very early in its history. By 1962, an influential study had set the search for extraterrestrial life as the prime goal of NASA's space biology program (National Academy of Sciences, 1962, pp. 9-2, 3). Culminating with the biology experiments on the Viking landers in 1976, NASA vigorously prosecuted this complex task, with the main result that life was not detected on Mars (Klein, 1977, p. 4677; Ezell and Ezell, 1983).

While Viking was still on the drawing boards, NASA began to contemplate an even more daring venture. This was the detection of interstellar communication — a "search for extraterrestrial intelligence" (SETI). The SETI project, as it was called in 1976, differed from its microbe-seeking cousin in that NASA had no realistic prospect of sending spacecraft to the planetary systems of other stars. A major assumption upon which NASA would build its SETI program was that interstellar travel by humans or any other intelligent beings was unlikely. Also in contrast to the Viking project, responsibility for SETI did not fall clearly into the province of the nation-

al space agency, NASA. Anyone opposed to SETI could argue that taxpayers' money should not be spent on this longest of long shots when so many other pressing needs existed. Moreover, the idea of extraterrestrial intelligence sometimes conjured up visions of horrific science fiction scenarios, from H. G. Wells' *War of The Worlds* to the movie *Alien*. A few members of Congress have shared this popular-culture concept of visiting extra-terrestrials, which was reinforced by contentious debates about Unidentified Flying Objects. For these reasons and others, the advocacy of SETI within NASA would be an uphill, albeit colorful struggle.

SETI had its origins not in the space program, but in the recognition that celestial objects radiate energy in many regions of the electromagnetic spectrum besides optical wavelengths. Two academic physicists — Giuseppe Cocconi and Philip Morrison at Cornell University — published the first theoretical paper on a strategy for interstellar communication. Frank Drake, then a young radio astronomer at the National Radio Astronomy Observatory (NRAO) in Green Bank, West Virginia, carried out the first search, which he dubbed "Project Ozma." Cocconi and Morrison were contemplating the possibility of interstellar communication via gamma rays, but arrived instead at the conclusion that the radio region of the spectrum was optimal for this purpose. In a landmark paper that initiated modern interest in SETI, they suggested the 21-cm emission line of hydrogen — the most abundant element in the cosmos — as an obvious wavelength to examine in a search for intelligent transmissions (Cocconi and Morrison, 1959).

Drake had independently come to the same conclusion. With the support of NRAO's first Director, Otto Struve, he was working on the equipment for the Ozma search when Cocconi

and Morrison's article appeared in the journal *Nature*. Drake's ensuing 200-hour look at the stars Tau Ceti and Epsilon Eridani revealed no evidence of ETI's (Drake, 1961), but his search, together with the paper of Cocconi and Morrison, raised many questions, fired the imagination of the public, and caused other scientists to declare (or develop) interest in the quest.

The events of 1959-60 inspired a meeting in November, 1961 at the NRAO in Green Bank. This informal conference, sponsored by the National Academy of Sciences, is best known for giving birth to the "Drake equation," a formula for estimating the number of communicative civilizations in the Galaxy (Cameron, 1963, pp. 288, 310). Codifying in concise form the attempts of Harvard astronomer Harlow Shapley (1958) and others to determine the probable number of extraterrestrial civilizations by calculating astronomical, biological, and social factors, the Drake equation became the perennial guide for SETI discussions. Few of its parameters could be evaluated reliably, however. Never before in the history of science had an equation been formulated that yielded answers differing by eight orders of magnitude, depending upon the values assigned its parameters. If skeptics saw this latitude as a lack of rigor, others saw it as a challenge to devise apposite experiments. For both optimist and pessimist, the Drake equation became a means for arguing a point basic to SETI — whether or not communicative civilizations exist in abundance in the universe. The participants in the Green Bank conference spent some time considering the intelligence of dolphins, to which they felt such an affinity that they called themselves the "Order of the Dolphins." The social history of the Drake equation that they formulated is considered more fully later.

No representative of NASA attended the

Green Bank meeting, although some of the papers presented there were later edited by an astronomer affiliated with NASA, A.G.W. Cameron (Cameron, 1963). At that time, NASA had a greater interest in the immediate prospects offered by space exploration. This is evidenced by topics considered at a meeting in 1963 on "Current Aspects of Exobiology," held at the Jet Propulsion Laboratory (a NASA-funded agency administered by the California Institute of Technology). Its proceedings (Mamikunian and Briggs, 1964) were filled with discussions of methods of detecting life on Mars. At the same time, the inclusion of a paper by Drake on interstellar communication showed that the search for intelligence beyond the solar system might be considered a more advanced part of the then-nascent discipline of exobiology.

During the 1960's, NASA promoted studies in planetary biology. The Agency solved practical problems such as biological contamination of the planets by spacecraft or back-contamination of the Earth by planetary microorganisms (Pittendrigh and others, 1966), but did not pursue interstellar communication. The situation in the Soviet Union provides an interesting comparison. Soviet SETI was stimulated by the Soviet space program. Well aware of the paper by Cocconi and Morrison, radio astronomer Joseph Shklovskii at the Sternberg Astronomical Institute of Moscow University wrote a book in 1961 on extraterrestrial intelligence for the fifth anniversary of the launching of Sputnik. Published in 1962, *Vselennaia, Zhizn, Razum (Universe, Life, Mind)* received wide circulation in the West after Carl Sagan revised and expanded it for an English edition (Shklovskii and Sagan, 1966). Shklovskii's volume stimulated his young colleague Nicolai Kardashev to organize a conference on Extraterrestrial Civilizations, which was held in 1964 at the Byurakan Observatory in Soviet Armenia

(Tovmasyan, 1965). The English translation of Shklovskii's work stimulated interest in SETI among NASA scientists (Swift, 1990, p. 251). Radio astronomy and the space age thus worked together to promote interest in interstellar communication.

NASA's first publicly expressed interest in SETI in 1970 derived from a confluence of factors. Since 1960, experiments had been carried out on the origin of life in our planetary system at the Life Sciences Laboratory of NASA's Ames Research Center in California. Ames had been home to the team in planetary exobiology that undertook the search for microbial life on Mars. Ames also had responsibility for space medicine. In 1969, John Billingham, a physician who headed the Biotechnology Division, convinced Ames' new director, Hans Mark, to undertake a mini-study of the propriety of SETI within NASA. This resulted in NASA's first series of lectures on Interstellar Communication in 1970 (Ponnamperuma and Cameron, 1974) and, more importantly, in an internal study of its feasibility, conducted at Ames.

Following the success of the lectures of 1970, a more ambitious "design study of a system for detecting extraterrestrial intelligent life," known as Project Cyclops, was organized by Billingham in 1971. The study was not a major NASA project. Rather, it was part of a summer faculty fellowship program in engineering systems design sponsored by the Ames Research Center and Stanford University. The study addressed details of antenna elements, receiver systems, and signal processing, as well as more general problems about the probability of life in the universe and search strategies. A key player was Bernard Oliver, a distinguished scientist and electrical engineer, Vice President of Hewlett-Packard, and member of the Order of the Dolphins. Oliver wrote most of the final

Cyclops report (Oliver and Billingham, 1972, revised 1973). He and his colleagues envisioned a detector consisting of an orchard of perhaps one thousand 100-meter antennas covering a total area some 10 km in diameter (Figure 5). Cyclops was nothing if not ambitious. But considered as a phased additive array of connected antennas, the system had the capability of starting out small and building further if the first few antennas detected no signals.

The Cyclops report holds an ambivalent place in SETI history. On the one hand, it recommended “establish[ing] the search for extraterrestrial intelligent life as an ongoing part of the total NASA space program.” On the other hand, the full project it suggested, which demanded 6-10 billion dollars over 10-15 years, was much too ambitious for NASA funding. (However many critics did not understand that the project began with one antenna, at modest cost, and could have succeeded with one antenna in a short time. Most critics jumped to the conclusion that the cost was necessarily 6-10 billion dollars, and did not read the details about the incremental approach.) Nevertheless, Oliver and Billingham’s study served as an indication of what could be done, demonstrated the technical feasibility of interstellar communication, and provided a benchmark against which smaller programs could be measured. In 1972, Oliver presented the results of the Cyclops study at a meeting of the International Academy of Astronautics and brought the project to international attention.

From Cyclops to the HRMS (High Resolution Microwave Survey)

Despite the recommendation of the Cyclops report, NASA funding for SETI

was limited during the early 1970’s to workshops and conferences. These served the purposes of keeping interest alive and refining technique. Most important was a series of six Workshops on Interstellar Communication held in 1975-76 under the chairmanship of Philip Morrison (Morrison, Billingham, and Wolfe, 1977). These workshops, and three offshoots dedicated to the detection of planets and the study of the evolution of technological civilizations, proved to be a landmark in SETI history and a critical stimulus to enlisting support by the wider scientific community. Having considered interstellar travel, robot probes, and electromagnetic signals, the report confirmed that radio signals would be the optimum method for interstellar communication. It also recognized that the search for signals had to be limited in direction, or frequen-



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cy, or both. Although no consensus was reached on a search strategy, the report (in a paper by Charles Seeger) gave the first public discussion of a possible bimodal method for the search – a detailed look at selected target stars in parallel with a broad-brush all-sky survey (Seeger, 1977, p. 85). The assumptions behind each strategy would be the subject of much further discussion. The studies of 1975-76 also revived interest in the possible existence of extrasolar planetary systems and stimulated a NASA study of a method for detecting them (Black, 1980).

Just as important as the conceptual recommendations of the Morrison workshops was an administrative conclusion: the participants agreed that “it is both timely and feasible to begin a serious search for extraterrestrial intelligence.” They also argued that the search fell under NASA’s mandate:

It is particularly appropriate for NASA to take the lead in the early activities of a SETI program. SETI is an exploration of the Cosmos, clearly within the intent of legislation that established NASA in 1958. SETI overlaps and is synergistic with long-term NASA programs in space astronomy, exobiology, deep space communication and planetary science. NASA is qualified technically, administratively, and practically to develop a national SETI strategy based on thoughtful interaction with both the scientific community and beyond to broader constituencies..

(Morrison, Billingham, and Wolfe, 1977, p. 34).

Accordingly, NASA established a small SETI Program Office at the Ames Research Center in 1976. Headed by John Billingham, this was the first institutionalization of SETI within NASA. It was made possible by the continuing support of Hans Mark, Director of the

Ames Research Center, and by Harold Klein, Director of Life Sciences at Ames.

The mention of the Deep Space Network and planetary science in the Morrison report signalled the interest of the Jet Propulsion Laboratory (JPL) and the support of its prospective Director, Bruce Murray. At both JPL and Ames, the innovative SETI programs stemmed from the personal interest and support of the new directors. The interest at JPL developed naturally, since JPL ran the Deep Space Network of radio telescopes and had expertise in the radio astronomy needed for SETI. But the crucial ingredient was Murray, who as Professor of Planetary Science at Caltech had participated in the Morrison Workshop on Interstellar Communication (April 1975) which dealt with detecting planets (Murray and Gulkis, 1978). Murray championed the all-sky survey for SETI against the skepticism of the Ames group, which pushed for a more traditional targeted search. In the fourth workshop in December of 1975, Billingham and Seeger had presented a paper on “Ames-JPL Plans” for a detector. By 1977, JPL had a SETI office, headed by Robert Edelson (Murray, Gulkis and Edelson, 1978). After some initial conflict, an Ames-JPL partnership emerged that would become a major feature of NASA’s formal SETI program.

Outlook for Space, prepared in 1976 by contributors from all the NASA centers to guide NASA’s thinking for the next 25 years, viewed investigations into the origin and existence of life, whether microbial or intelligent, as an important part of NASA’s space objectives through the end of the century (NASA, 1976, pp. 38, 145-9). The possibility of increasing the scope of NASA’s exobiology program from the search for microorganisms within the solar system to the search for extrasolar planetary systems and radio signals from an ETI —

from the confines of the solar system to the entire cosmos – was a breathtaking leap. By 1976 – the year of the Viking landers and the bicentennial of the United States – SETI was continuing to gain respectability in NASA.

Propelled by the Morrison workshops and emboldened by *Outlook for Space*, NASA sought to devise a program that might be funded. SETI would be significantly unlike most NASA endeavors. It would have no spacecraft, no launch risks, no possibility of equipment failure in space. Political and economic realities dictated that SETI would have no Cyclops system with a vast collecting area. Instead, the embryonic program would use

existing radio telescopes to which would be attached specialized detectors and signal processing apparatus whose construction would be the main objective of the funding. The proposed total cost of the SETI program as calculated in the late 1970's, including five years of research and development and ten years of operation, would be about 100 million dollars — some ten percent of the billion-dollar Viking project but roughly equal to the cost of Viking's biological experiments.

In June 1979, with the possibility of significant funding on the horizon, NASA sponsored a major conference at the Ames Research Center on "Life in the Universe." With the

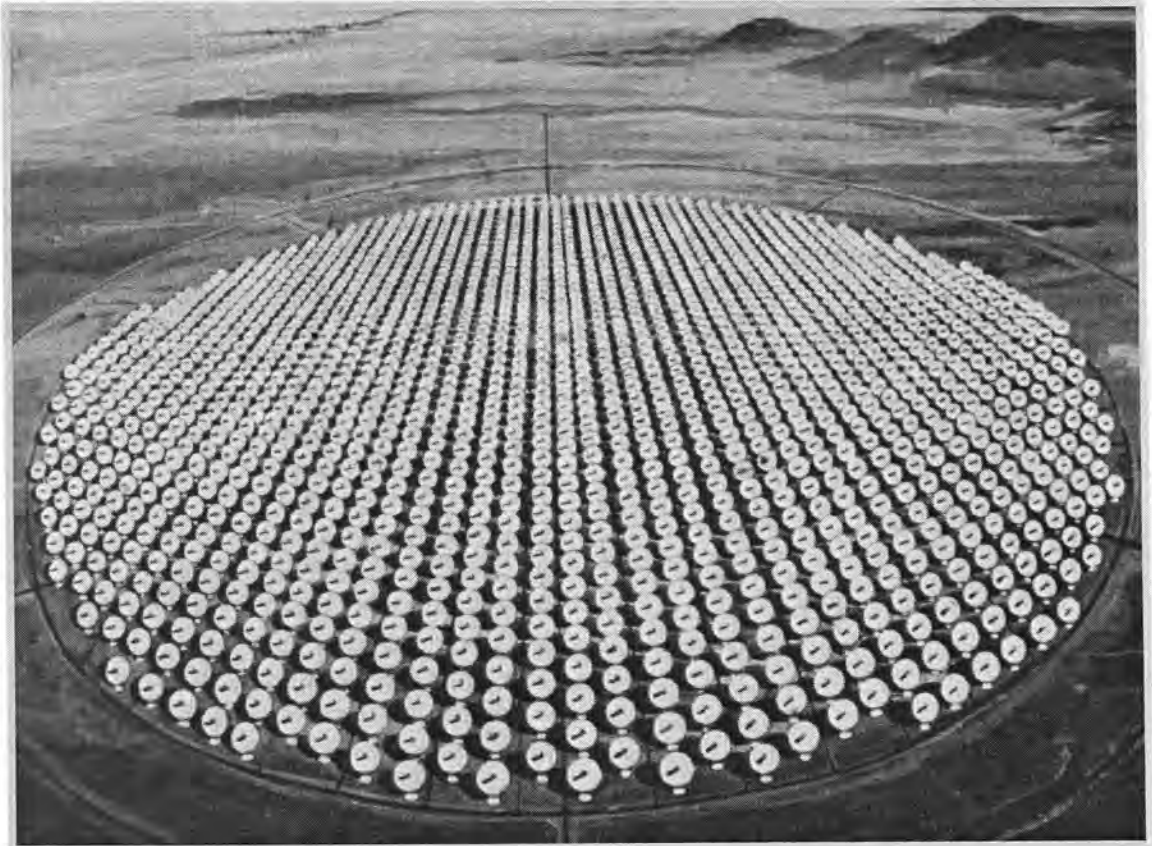


Figure 5. An artist's conception of the full implementation of the Cyclops array.

impetus provided by the Morrison workshops, NASA by this time had formally adopted a bimodal search strategy — a combined targeted search and sky survey approach that made sense scientifically and satisfied the desires of both JPL and Ames to work on the project. Scientists at Ames and JPL co-authored one of the papers given at the 1979 Conference, the first to lay out the NASA program in detail (Wolfe and others, 1981). Referring to their “modest but wide ranging exploratory program,” the authors described a ten-year effort “using existing radio telescopes and advanced electronic systems with the objective of trying to detect the presence of just one signal generated by another intelligent species, if such exists.” (The emphasis on detection is significant — NASA was not prepared to transmit). JPL would undertake Murray’s all-sky survey, at frequencies from 1 to 10 GHz, while Ames would concentrate with more sensitivity on the “targeted search” among some 700+ stars within 25 parsecs (80 light years) of Earth.

The Ames-JPL group characterized the concept of intelligent life as an hypothesis widely held in the scientific community. The hypothesis rested on two postulates: 1) that life is a natural consequence of physical laws acting in appropriate environments, and 2) that physical processes that we can observe in operation on Earth will operate in similar ways elsewhere. The group also ventured a conjecture that it called a ‘matter of practicality’: some ETI’s would “provid[e] an electromagnetic signal we can recognize (Wolfe and others, 1981, p. 392).” They pointed out that many searches had been undertaken with comparatively primitive data-processing systems. They anticipated a ten-millionfold increase in capability over the sum of all previous searches and recommended a major effort to develop the necessary equipment. The key instrument, known as the Multi-Channel Spectrum Analyzer (MCSA), and its software algorithms

were the heart of the system, the means by which the “cosmic haystack” could be searched for its “needle” (a favorite metaphor employed by NASA). By 1979, NASA had a detailed idea for a coherent SETI program, but not much money to carry it out.

During the 1970’s, NASA studied the SETI problem; during the 1980’s, the Agency intended to implement the recommendations of the studies. Studies and refinements would continue, notably in meetings during 1980 and 1981 of a SETI Science Working Group (SSWG) composed of academic radio astronomers and headed by John Wolfe of Ames and Sam Gulkis of JPL (Drake, Wolfe, and Seeger, 1983). To convert concepts and discussion into hardware and software required money. After detailed Congressional hearings in September, 1978, the Subcommittee on Space Science and Applications of the House Committee on Science and Technology supported NASA’s proposal to initiate a SETI program (U.S. Congress, 1979). However, the House and Senate Appropriations Committees elected not to appropriate any funds. That happened in 1979, after the program received one of Senator William Proxmire’s notorious Golden Fleece Awards for “unnecessary expenditures in the Federal Government.”

NASA continued to fund SETI at a subsistence level after 1979 until thwarted again by Senator Proxmire, who killed all funding for SETI for Fiscal Year 1982 through an amendment process. Nevertheless, NASA decided to return to Congress for full funding in fiscal year 1983. The Agency was emboldened by a report by the National Academy of Sciences that recommended SETI as one of seven moderate programs that NASA should implement (National Research Council, 1982). After maneuvers that included a friendly chat between Cornell astronomer Carl Sagan and

Senator Proxmire, and again with the backing of Hans Mark, then Deputy Administrator of NASA, SETI funding was restored for fiscal year 1983 at the level of 1.5 million dollars.

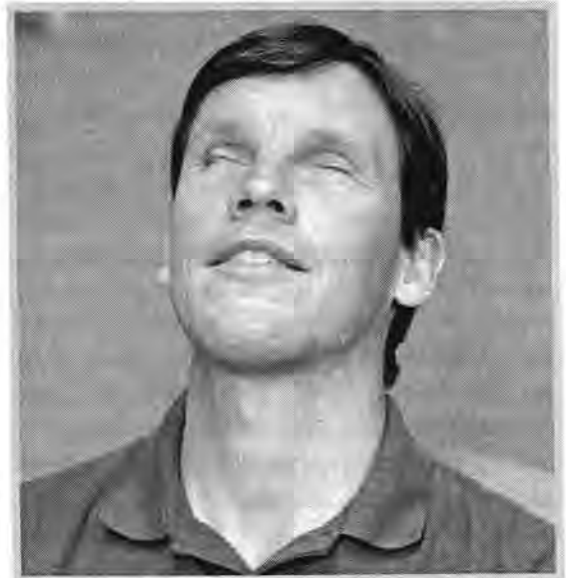
With funding in hand, NASA's Ames and JPL researchers embarked on an intensive program, known first as the Microwave Observing Project (MOP), then as the High Resolution Microwave Survey (HRMS, since October 1, 1992). Initial development built the instrumentation necessary for a systematic search for intelligent life, with Ames concentrating on the MCSA and JPL on an analogous Wide Band Spectrum Analyzer (WBSA). Between 1983 and 1987 the Centers spent a combined total of about \$1.5 million per year. The first field tests of the prototype MCSA were conducted in 1985 (Tarter, 1988) and survey strategies were developed and tested at Goldstone (Klein and others, 1992). At NASA Headquarters, Donald DeVincenzi served as the SETI Program Manager from 1983 to 1986, followed by Miss Lynn Griffiths who filled that position until in late 1988, when Gary Coulter became the Program Manager. At JPL, Michael Klein (who had taken over from Edelson in 1981) forged a collaboration with JPL's Telecommunications and Data Acquisition (TDA) office to design and build an engineering development model of the WBSA (Quirck and others, 1988). SETI inspired the design of the model, but the multi-mission Deep Space Network, which is operated by the TDA, would share in its use.

In 1990 SETI took on the status of an approved NASA project, now beyond the "Research and Development" phase, and began "Final Development and Operations" to be completed by the year 2001 at a total cost of \$108 million. Administratively, SETI had gone from a small office within a Division at Ames in 1976 to a large group in two Centers with a total staff of about 100 in 1991.

Fiscally, its annual budget had risen from a few hundreds of thousands of dollars annually in the early 1970's to over ten million annually in the 1990's. Conceptually, its strategy had been honed and reduced to technically and politically realistic proportions since the visionary Cyclops days.

On the national scene, these and earlier developments (including increasing recognition of SETI's potential by other scientists) moved SETI into the ranks of soundly established sciences. The Astronomy Survey Committee of the National Academy of Sciences compared SETI with other astronomical research with the words "... intelligent organisms are as much a part of the universe as stars and galaxies ...", concluding with a statement that precisely defines the challenge to participants in this workshop:

"It is hard to imagine a more exciting astronomical discovery or one that would have greater impact on human perceptions



Kent Cullers, SETI signal processing specialist.

than the detection of extraterrestrial intelligence." (National Research Council, 1982, Field Report, page 150)

The scientific legitimacy of SETI was underscored by the same committee with new members nine years later:

"The search for extraterrestrial intelligence (SETI) involves, in part, astronomical techniques and is endorsed by the Committee as a significant scientific enterprise." (National Research Council, 1991, Bahcall Report, page 62)

Biologists, a group somewhat more skeptical of the abundance of intelligent life elsewhere than astronomers, nevertheless concurred in a separate recent opinion for the National Academy of Sciences that SETI should be pursued. Implicitly stating a view that life and intelligence are likely enough to arise elsewhere to warrant a detection effort, the Committee on Planetary Biology and Chemical Evolution of the Space Studies Board devoted some 20 pages of its report to approaches to detecting extrasolar biologies, many of which center on SETI (National Research Council, 1991). Despite such endorsements, SETI remained vulnerable to potshots from Congress, as shown by the *Congressional Record* of June 18, 1990 containing quotes from supermarket tabloids and other poorly-informed negative comments.

International scientific support for SETI also grew steadily. A few months after the Ames Conference in 1979, NASA unveiled its bimodal search strategy for the members of the International Astronomical Union during a special session attended by SETI pioneers Joseph Shklovskii and Philip Morrison (Gulkis, Olsen, and Tarter, 1980, p. 93). NASA reported progress on field tests of prototype instruments and other developments to

the Union's meetings in 1984 (Papagiannis, 1985), in 1987 (Marx, 1988), and in 1990 (Heidmann and Klein, 1991). Particularly important was the software team's work on "detection algorithms," the means of distinguishing a signal from myriad background radio noises (Cullers, 1985). Reflecting the interdisciplinary nature of the subject, other societies also took an active interest in SETI and its offshoots, notably the Planetary Society, the International Society for the Study of the Origin of Life (ISSOL) and the International Academy of Astronautics (IAA). In 1982 the IAA appointed a SETI Committee (Bioastronomy), the first such committee established by an international space society. Small-scale SETI observations were carried out worldwide (Tarter, 1985). Billingham cautiously characterized the new discipline in 1981 as having "reasonable maturity and respectability."

NASA was also studying related projects, including a search for extrasolar planets (Black and Brunk, 1986), the origins and distributions of biogenic compounds (Wood and Chang, 1985), the origins of life (Hartman, Lawless, and Morrison, 1985), events in space that might affect the evolution of complex life on Earth (Milne and others, 1985), and exobiology in Earth orbit (Defrees and others, 1989). Most of NASA's SETI work remained focused, however, on the detector hardware and the associated algorithms. The question, how to design the ultimate mousetrap for an information-bearing signal, raised the familiar difficulties and ambiguities of SETI and required for its answer many assumptions and informed conjectures about manifestations of distant intelligent life. It would not be easy (or perhaps even possible) to make much sense of a detected signal. And if the mouse-trap worked, NASA would confront the problems of when and how to inform the public. While the Agency wrestled with these prob-

lems, the SETI Committee of the International Academy of Astronautics devised a set of suggestions, known collectively as the “post-detection protocols,” for confirming receipt of the first signals and disseminating information about them (Tarter and Michaud, 1990; see also Chapter 4).

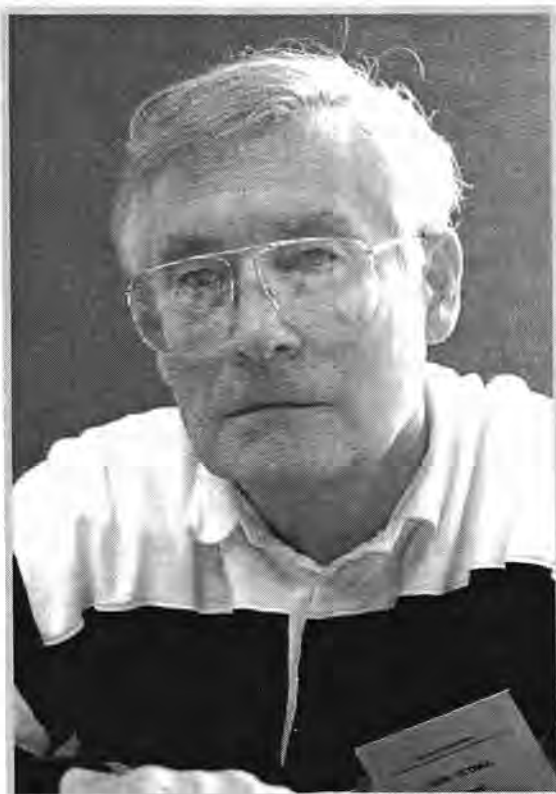
The Implications of Detecting a Signal

Almost from the beginning of its work in Exobiology, NASA has been sensitive to the philosophical implications and humanistic appeal of the field. The National Academy of Sciences’ *Report* of 1962 (pp. 9-2, 3) observed:

[Exobiology] has a far greater philosophic appeal than has been fashionable in scientific discussion for some time. But it is not since Darwin – and before him Copernicus – that science has had the opportunity for so great an impact on man’s understanding of man. The scientific question at stake in exobiology is, in the opinion of many, the most exciting, challenging, and profound issue, not only of this century but of the whole naturalistic movement that has characterized the history of Western thought for three hundred years. What is at stake is the chance to gain a new perspective on man’s place in nature, a new level of discussion on the meaning and nature of life.

In this statement, the Academy had in view the search for life in the solar system; the same attitude has pervaded the NASA SETI community as well.

Recognition of the philosophical implications of SETI research prompted concern about the cultural impacts of its outcome. Morrison’s workshops of 1975-76 identified “the impact of conducting a search, and the impact of success or failure in terms of national, international, social and environmental considera-



Karl Guthke, Professor of German Art and History at Harvard University.

tions” as a question worth pursuing (Morrison, Billingham, and Wolfe, 1977, pp. ix, 8). The workshop members recommended the question to their successors, since it lay beyond their scope and competence. They wrote: “The question deserves... the serious and prolonged attention of many professionals from a wide range of disciplines — anthropologists, artists, lawyers, politicians, philosophers, theologians — even more than that, the concern of all thoughtful persons, whether specialists or not.” Another clear signal of sensitivity to cultural impact in the Morrison workshops was the distinction made between the *search* for extraterrestrial intelligence (SETI) and *communication* with extraterrestrial intelligence (CETI). In a concession to those who urged caution about the implica-

tions of communication with advanced civilizations (as did the British radio astronomer Martin Ryle), the report explicitly distinguished between SETI (searching) as proposed by NASA and CETI (broadcasting or two-way communication) as enunciated by the Soviet Union in 1971 (Sagan, 1973a).

Stronger reassurance to some was given in the foreword to the report of the Morrison workshops by theologian Theodore Hesburgh, then president of Notre Dame University. He stated his belief that "this proposed search for extraterrestrial intelligence (SETI) is also a search for knowing and understanding God through his works." Other theologians, however, have been more cautious about the implications of extraterrestrials for theology (Sullivan, 1966, Chapt. 19). They remind us that our species is very diverse in cultural and intellectual background and that we will not be able to react in one voice to any discovery of ETI. Whatever the uncertainties about the outcome, however, SETI broadens the context of philosophical thinking and adds a new puzzle to questions of science and society.

During its brief modern history, SETI has been viewed in many ways. Its pursuits and assumptions have been seen as science (by most of its practitioners), as exploration (Billingham, 1991), as a test of the "biophysical cosmology" (Dick, 1989, 1991), as a thought system or "myth" in the broad positive sense of the term (Guthke, 1990), and as pseudoscience by a few skeptics (Tipler, 1987). These divergent interpretations notwithstanding, NASA chose to begin its SETI observing program on the 500th recurrence of the anniversary of Columbus' discovery of America. The inaugural date, the feeling of some that SETI is more exploration than science, and uncertainty about whether success would have beneficial consequences, all make the reference to the voyages of

Columbus appropriate, challenging, and perhaps instructive.

The Broad Sociological Context of SETI

The Drake Equation and the Fermi Paradox

While preparing the agenda for the meeting of the Order of the Dolphins at Green Bank in November 1961, Frank Drake realized that each of the topics he had scheduled could be presented by a factor in an equation that gave the likelihood of the existence of detectable extraterrestrial intelligence. This expression, now known as the Drake Equation, links the number N of communicative societies existing in our galaxy at any time with the rate of formation of such societies and their average length of technological life:

$$N = R^* f_p n_e f_l f_i f_c L$$

The factors on the right are R^* , the rate of star formation in the Milky Way Galaxy; f_p , the fraction of stars that have planets; n_e , the number of planets per star suitable for life; f_l , the fraction of these planets on which life develops; f_i , the fraction of life-bearing planets with intelligent life; f_c , the fraction of intelligent cultures that develop detectable radio communication; and L , the average time spent by civilizations in a communicative state (Cameron, 1963, p. 188). The product of the factors before L gives the rate of formation of communicative civilizations. Hence N represents the number of detectable civilizations in the galaxy at any time.

The Drake Equation may appear to be an odd way of organizing thought. The values of some of its parameters, like R^* , f_p , and n_e , are commonly estimated to within a factor of ten or a hundred. The values of the others, f_i , f_c , and L , are purely conjectural. The number of intelligent societies in our galaxy so calculated ranges from 500 million (Wertz, 1976) to one (Tipler, 1980) and even to an unflattering less-than-one (Hart, 1982). A recent estimate of the Drake parameters makes R^* about 10/year, f_p about 0.5, n_e and f_l about 1, and f_i and f_c anywhere between 0 and 1; to L it

assigns the astounding range of 10 to 10^{10} years (Jones, 1991). People who calculate N must bring to the work more than the facts. Both in its form and its factors, the Drake Equation causes SETI analysts to rethink their general underlying assumptions of the scientific enterprise and their personal world views.

As to form, the Drake Equation might be considered more sociological than mathematical. The equation generates a number N that lies between one and 500 million from a product of several factors, many of whose values are uncertain. The exercise might therefore fall under the practice of scholastic philosophers called *ignotum per ignotius*, the explanation of the unknown by the more unknown. The form is science; the content, speculation; the purpose, a convenient organization of the unknown.

Evaluations of the factors f_l , f_i , f_c , and L relate to the hopes, fears, and fancies of the calculators. The first factor, representing the probability that an Earth-like planet will develop life within a few billion years of its formation, is occasionally put at close to zero by some biologists (Ornstein, 1982, p. 31). Most exobiologists estimate the chance at close to one, and supporters of SETI also bet

on a number near unity (Oliver, 1975, p. 361). There is much invested here: $f_l = 1$ is necessary for a high probability of detection of ETI's; $f_l = 0$ insures that no ETI signal will be found. Much the same can be said about guesses about f_i . Some biologists come down near zero, and some astronomers near one (Tipler, 1980, p. 267). A few skeptics (Tipler, 1981, p. 286) claim that the high estimate of the probability that life will develop intelligence smacks of teleology, of a directed goal for evolution. Others disagree profoundly (Sagan and Newman, 1983).

The point can be made more strongly by reference to the so-called "principle of mediocrity," according to which intelligent life should be widespread in the galaxy since it is present on our otherwise undistinguished planet orbiting a typical star in a remote arm of the galaxy. Besides making intelligence commonplace, the principle has the further merit, in the minds of its proposers, of accomplishing the final deprovincialization of the human world view. The displacement of mankind from the center of the universe, begun by Copernicus and followed by the removal of our species from a central role in the cosmic story by the theory of evolution, is now completed by the realization that intelligence is likely to be common in the galaxy (Sagan and Newman, 1983, p. 113). The principle of mediocrity may appear to express modesty and even humility. By making intelligence a natural and inevitable feature of life in the universe, it raises the quality most prized by humans to a norm of the universe. In contrast, the mouthpiece of the aardvark, the sex dance of the partridge, or the coloration of a butterfly are unique.

The anthropomorphism of the principle of mediocrity is carried further by estimates of

the value of f_c , the fraction of planets bearing intelligent life that develop communicative technological civilizations. Again, interpreters of Drake's equation sympathetic to SETI assume f_c close to one, implying that intelligent life usually gravitates toward a technological society (Freeman and Lampton, 1975). (Biologists seem disinclined to speculate about the value of f_c .) A technological society eventually will hit upon signalling, radio, radar, and television. The radiations from these devices, whether leaked or beamed intentionally, are the quarry of SETI. Electromagnetic civilizations, which are likely to be much older than ours, will have passed through much the same intellectual evolution that we are undergoing to produce *Homo technicus*.

Do these older civilizations yet endure? We come to L , the average length of life of a communicative society. Here even the boldest speculators admit perplexity. Nonetheless, they reason about the fates of our putative predecessors by expanding our late impasses over nuclear disarmament and proliferation to fill the universe. Earlier technological societies may have travelled similar paths; if they learned to live with or destroy their weapons of mass annihilation, they might thrive for a billion years. If not, they might exterminate themselves, as we may do, within a thousand years of their first electronic experiments (Oliver, 1975, pp. 365-6; Sagan and Newman, 1983, p. 119). This extrapolation of possible human futures to the whole universe makes "man the measure of all things" in a way that would have astonished Protagoras.

In an idle moment at Los Alamos, Enrico Fermi asked "where are they?" This question, now known as the Fermi Paradox, asks why, if intelligent extraterrestrials exist, we do not see the evidence of their existence about us

(Finney and Jones, 1985, pp. 299-300). The connection in this apparent *non-sequitur* is supplied by ascribing to ETI's the same impulse to colonization felt by humans. As Frank Tipler, one of the leading reasoners about the Fermi Paradox, puts the point: "It is difficult to construct a plausible scenario whereby an intelligent species develops and retains for centuries an interest in interstellar communication together with the technology to engage in it, and yet does not attempt interstellar travel" (Tipler, 1980, p. 274). In Tipler's vision of colonization, the explorers proceed by proxy, using machines capable of exploiting the resources of distant planets and asteroids to reproduce themselves and spread throughout the galaxy. The plethora of self-reproducing machines would not take long to reach every part of the galaxy, including our backwater. Tipler calculates about 30 million years, at a moderate and sustainable pace. Others whose envisioned robots march at a faster tempo arrive at under 10 million years (Hart, 1975, pp. 128-35). If these conjectures reflect reality, there has been time for many visits to Earth by self-reproducing machines. Why have we no relics of the journeys of these space-farers? Why did they not colonize our planet?

Two answers may be given. According to Tipler and Hart, they did not come because they never did exist (Tipler, 1981, p. 290). According to Shklovskii's late pessimistic view, they did not come because they did not survive their political systems long enough to travel (cf. Papagiannis, 1978, pp. 277-81). According to Oliver (1992), they have not come because colonization either directly or by proxy is far too costly in time and energy, because of the enormous distances involved, to recommend itself to any intelligent society. According to Ball (1973), they have declared the Earth off limits, either to isolate what they reckon to be a potentially-troublesome species

or to preserve our curious species as their "zoo." We are assured, in any case, of the benignity of the visitors. An exploring civilization would have to hold together for at least 10 million years, and perhaps as long as a billion, to complete a reconnaissance of the galaxy. To hold together for so long, a technological society would have had to lay aside all forms of aggression that could lead to planetary annihilation. "Their mere existence makes it highly likely that their intentions are benign and their sensitivities about societies at our level of technological adolescence delicate" (Sagan and Newman, 1983, p. 120).

Experimentalism and its Politics

The foregoing collection of speculations omits a point of view that continues to gain strength among the technical staff of NASA's SETI project. "Stop speculating and start looking" was the answer they gave to debates about the existence of extraterrestrials. Fermi, they said, had asked a question that should be answered experimentally, not posed as a paradox to be debated endlessly. Cocconi and Morrison's proposal to look for electromagnetic waves from extraterrestrial civilizations and Drake's pioneering search for a radio signal from nearby stars pointed toward a practical experimental approach to the old question of whether or not we are alone. If our civilization could be detected great distances away by the radio signals we send out into space, so the experimentalist logic went, a search through the radio spectrum should enable us to detect extraterrestrial civilizations like our own.

Proponents of this experimental approach present a chain of plausible assumptions to justify their call for a search. There are tens of billions of long-lived, stable stars like our own in the Galaxy, which may have planets with highly evolved, intelligent life. The no-

nonsense experimentalists infer that other intelligences will have discovered and used radio, just as we do. All radio-using civilizations in the galaxy probably will know the frequencies for communicating most economically over long distances. These frequencies, as portions of the radio spectrum with the least background noise, are therefore the most promising ones to examine for signals intentionally sent by ETI's.

By proposing a passive strategy of searching for a signal from an extraterrestrial source, rather than an active strategy of transmitting into space in hopes of attracting a reply, the experimentalists simplified the problem and also accommodated a desire for results in the short term.

Not until the computer revolution, however, was it possible to contemplate mounting a systematic search that would not require major funding for massive arrays of antennas. The new concept coupled existing antennas of moderate size with intensive data processing equipment, significantly reducing total cost. In addition to winning over opponents who might otherwise have considered the enterprise too expensive, a computer-aided radio search gained the support of many mathematicians and computer scientists who found it an interesting intellectual challenge, rich with prospects for technological spin-offs, irrespective of the likelihood of success at detection.

As search strategies were refined and signal-processing equipment was developed during the 1980's, SETI proponents sought to avoid promising too much by explicitly anticipating the significance of possible outcomes from their search. While admitting that the detection of a signal from an extraterrestrial intelligence would be a stunning discovery as well as an affirmation of their search logic, they cautioned that any failure or series of failures

to detect an intelligent signal from the stars would not prove that we are alone in the Galaxy. Instead, they stressed that it would only discredit one or more of their assumptions – such as their choice of frequencies for the search or the more basic premise that other intelligences communicate by radio. SETI scientists further cautioned that civilizations that might be using radio to communicate could be too thinly spread through the Galaxy to give the first search much chance of success. Consequently, they said, the quest would probably have to be expanded incrementally by searching an ever widening range of frequencies with instruments of increasing sensitivity before they could begin to comment on the presence or absence of other civilizations in a radio-communication phase.

Although SETI scientists could not promise an immediate answer to their central question, they did raise the prospect of the serendipitous discovery of currently unknown radio astronomical phenomena, citing the numerous astronomical findings that have resulted from new instrumentation that made possible the intensive examination of hitherto neglected portions of the electromagnetic spectrum.

The politically and intellectually viable experimentalist opinion has an antiseptic character made possible by the progressive gains in the battle over funding for SETI. Arguments over first principles and cosmic visions, which attracted attention to the fledgling field and invested it with romantic interest, no longer seem pertinent. The funding sought is small by NASA standards, the spin-off in data analysis and in unanticipated astrophysical discoveries might be substantial, and the business has become (or is said to be) “science as usual,” at least among the younger experimentalists.

The current ascendance in scientific circles of

this carefully delimited experimental approach does not necessarily mean that historical analogues involving contacts and confrontations between terrestrial cultures can now be safely forgotten. However frequently SETI scientists may reiterate that they seek only to detect a radio signal from an extraterrestrial intelligence, not to engage in a physical encounter or even in two-way radio communication, analogic thinking based on contact between human societies persists. Although many of the analogies from previous human experiences that were central to the debate about extraterrestrials may be declared invalid within the narrow confines of a radio search, they are likely to resurface as more and more people come to realize that a systematic search is underway. They are especially likely to surface if the search reveals that we have intelligent neighbors in space. Hence it makes good sense to examine these analogies anew. Particularly in the light of SETI speculations about the likely technological superiority of any extraterrestrial civilization it may detect, analogies that feature the intellectual and social impact of the transmission of ideas between cultures and civilizations, as opposed to those that stress the physical consequences of face-to-face encounters, may bear special scrutiny. This scrutiny, at present and in the past, consists of speculation that divides into “catastrophic” and “millenarian” visions of the likely social consequences of a confirmed detection of an ETI. The thinking organized or inspired by the Drake equation, the Fermi paradox, and the experimentalists may seem sober in comparison with millenarian and catastrophic speculations largely unrestrained by mathematics.

Catastrophists and Millenarians

Sunny views of the unaggressive nature of extraterrestrials are evident in the report of

the Morrison workshops, *The Search for Extraterrestrial Intelligence*. That is not because its authors have not heard warning voices:

Some have worried that a message from an advanced society might make us lose faith in our own, might deprive us of the initiative to make new discoveries if it seems that there are others who may have made those discoveries already, or might have other negative consequences. But we point out that we are free to ignore an interstellar message if we find it offensive... The receipt and translation of a radio message from the depths of space seems to pose few dangers to mankind; instead it holds promise of philosophical and perhaps practical benefits for all of humanity.

Other imaginative and enthusiastic speculators foresee big technological gains, hints and leads of extraordinary value. They imagine too all sorts of scientific results, ranging from a valid picture of the past and the future of the Universe through theories of the fundamental particles to whole new biologies. Some conjecture that we might hear from near-immortals the views of distant and venerable thinkers on the deepest values of conscious beings and their societies! Perhaps we will forever become linked with a chain of rich cultures, a vast galactic network. Who can say? (Morrison, Billingham, and Wolfe, 1977, p. 8).

This statement summarizes and takes a position in a long-standing debate. The position, which we shall call "millenarian" after the old belief in a millennium before the final dissolution of the world, is that of commentators who believe that detection of an ETI civilization is likely to have profoundly beneficial consequences for humanity. Both the millenarian and its opposite, the "catastrophic" view, have been prominent throughout history. In earlier

ages, fears and hopes were conditioned by the challenge that extraterrestrial intelligence might offer to the most precious contemporary values. These values included humanity's perceived supremacy in God's grace and the more secular conviction that "rational man" is the measure of all things and is thereby destined or preconditioned for "happiness." Later, the Darwinian confidence that *Homo sapiens* represents the highest stage of development in the organic scheme of things, and the conviction that the human species has full control of our planet, were added to, or exchanged for, the earlier values.

The values just described survive in some form in segments of the living population, whose reactions it might therefore be possible to foresee. Also, history shows that, over the centuries, there have been certain constants in human behavior in response to the suggestion that "we are not alone." Equally helpful to our task may be the realization that, although we know much more about the physical universe than our predecessors, we have no more scientific evidence of the existence of extraterrestrial intelligence than did earlier people who speculated about the plurality of worlds.

Even the impersonal, indirect encounter by radio signals that SETI envisages does have analogues, of sorts, in earlier thinking. We do not have in mind the notion of Camille Flammarion, the foremost champion of the plurality of the worlds in the nineteenth century, who in his novel *La fin du monde* (1894) relied on psychic communication with the inhabitants of Mars. We think instead of the fear or hope raised about the philosophical impact of the existence of extraterrestrials, often presented not as putative but as a common-sense certainty. This was not a harmless intellectual diversion: it mattered to many whether humanity was, or was not, the apple of God's eye or the measure of all things.

Those historical parallels, as well as reasoning based on the contemporary scene, usually presuppose that "they," the extraterrestrials, are like "us," the human beings. Anthropomorphism cannot be avoided here. If ETI takes the shape and "mind" of a cloud (Fred Hoyle, *The Black Cloud*) or a planet (Stanislaw Lem, *Solaris*), we will lose our bearings altogether. If our "reason" (or psychology, or physics) does not prevail in civilizations elsewhere in space, how can we knowingly enter into contact with them? And how could we find it meaningful to conceive of "them," as we commonly do, as potential guardian angels or potential predators?

Some analysts think that, after the initial excitement, a signal from space would be treated with indifference by the population at large. In this view, profound consequences are unlikely (Regis, 1985, pp. 231 ff). Yet most commentators suppose that the detection of ETI's would have an impact so much more powerful than the hypothesis, however compelling, of their existence, as to dwarf the Copernican and the Darwinian Revolutions to the stature of "minor preludes" (L. W. Beck, *ibid.*, p. 15). "A positive result [of SETI] would be the most cataclysmic event in our entire intellectual history," according to Steven Jay Gould (1985, p. 413). It is not clear from this statement, however, whether Gould should be enrolled among the millenarians or the catastrophists.

Catastrophic Views

The Brookings Institution's report on *Proposed Studies on The Implications of Peaceful Space Activities for Human Affairs* observed that many societies, "sure of their place in the universe . . . have disintegrated when they had to associate with previously unfamiliar societies espousing different ideas

and different life ways" (Michael, 1960, p. 193). Nobel laureate Martin Ryle, Britain's Astronomer Royal and one of the founders of radio astronomy, addressing the International Astronomical Union in 1976, urged the governments of the world to refrain from any attempt to communicate with extraterrestrial civilizations or to make our existence known to them. He worried that if they knew about us, they might be encouraged to colonize our planet. *The Extraterrestrial Encyclopedia* warns that a colonialist invasion could occur by signal:

An alien race — under the guise of teaching and helping us join the cosmic community — might actually trick us into building devices that allow "Them" to conquer "Us"... The extraterrestrial "Trojan Horse" might arrive on radio waves. For example, the alien race could transmit the details of a computer-controlled biochemistry experiment that would secretly create their own life-forms here... (Angelo, 1991, p. 24).

Some catastrophists do not require even the mediated physical contact imagined by *The Extraterrestrial Encyclopedia*. An unequal exchange of information might be enough to paralyze us. Donald Goldsmith's reader, *The Quest for Extraterrestrial Life*, matter-of-factly offers a statement made by T. B. H. Kuiper and Mark Morris in the journal *Science* (1977): "Complete contact with a superior civilization (in which their store of knowledge is made available to us) would abort [our] further development through a 'culture shock' effect" (Goldsmith, 1980, p. 175).

George Wald, Nobel Prize winner in medicine, introduced this fear into the discussion of contact by signal. He could "conceive of no nightmare as terrifying as establishing such communication with a so-called superior. . .

technology in outer space.” The “whole human enterprise,” the “dignity and worth of man,” indeed his “meaning” would be in jeopardy the moment we became the passive recipients of superior knowledge of extraterrestrials, he prophesied at a symposium on *Life Beyond Earth and the Mind of Man* (Berendzen, 1973, pp. 17-19). Italian sociologist Roberto Pinotti returned to this “devastating” socio-anthropological scenario in the issue of *Acta Astronautica* carrying SETI’s post-detection protocol (Pinotti, 1990, pp. 109-15), and amplified it from mere scientific horror to a vision of global disorientation and “authority crisis.” Does it follow that Nigel Calder’s suggestion in 1978, that the purpose of microwave observation projects is not to achieve contact but to prove that “they are not there,” has some plausibility (Calder, 1978, p. 137)?

Contact “could be the beginning of the [cultural] end of humanity as we have known it,” as Michael Michaud paraphrases one negative interpretation (Bova and Preiss, 1990, p. 250).



Ben Finney, anthropologist at the University of Hawaii.

Still, humbling might be healthy and perhaps even necessary to survival (Shapley, 1958). It takes a very dour catastrophist to believe that mere knowledge of the existence of ETI could extinguish the manifold creative energies that have enabled mankind to overcome many sorts of menaces over the centuries.

Millennial Views

The gloom of the doomsayers is more than offset by the rapture of millennial enthusiasts. A signal “will change everything,” says the philosopher L. W. Beck, there will be “no limit to what ... we might learn about other creatures and ... ourselves.” (Regis, p. 15) “They” might reveal to us how to cure cancer or how to tame thermonuclear energy (*The New York Times*, Nov. 22, 1976, in Goldsmith, p. 269), or instruct us “in the techniques of stable world government” (Cameron, 1963, p. 1). At the symposium that occasioned George Wald’s catastrophist pronouncement, Richard Berendzen rose above mundane benefits to the spiritual ones:

And finally, it would perhaps provide us with the opportunity of joining what has been called the “galactic heritage.” Let us move beyond the obvious, trite matters of scientific or technological gain that contact might provide. It might also lead us to better social forms, possibly to ways to solve our environmental crises, and even improve our own social institutions. If these civilizations have been able to survive for thousands or tens of thousands of years, perhaps they could tell us how in their infancy they had been able to overcome these problems. Their very being would constitute an existence proof that such problems are not inevitably debilitating. Perhaps they could even tell us about new aesthetic forms, which would raise our consciousness, making life more enjoyable and rewarding. Thus the benefits could come

not merely in terms of technology and science, but also in the arts, literature, and humanities. And beyond that — and probably the most significant of all — contact would end our social and cultural isolation. (Berendzen 1973, 49-50)

Being effectively “immortal,” Frank Drake added in the issue of the *Technology Review* of June, 1976, “they” might teach us to achieve the same state of perfection (Drake, 1976, p. 22). Project Cyclops speaks of “the salvation of the human race” (NASA, 1971, p. 31).

But why would such a bonanza not bring about George Wald’s nightmare, the “degradation of the human enterprise”? Why would it not “threaten our egos, shatter our institutions, destroy our lives”? Physicist Philip Morrison (Morrison and Sagan, 1973, pp. 333-349; Berendzen, 1973, pp. 41-4), and anthropologist Ben Finney (1990, pp. 117-121) find reassurance in distance. They assume that most, if not all, ETI’s reside very far from the Earth. Information would trickle in over long periods. It would take decades or possibly centuries to decipher it; an exchange of greetings could take hundreds of years. At this pace, revolutions do not occur.

As for the supposition of demoralization by the mere notice of the existence of an ETI, the optimists observe that the human race, or much of it, already has powerful antibodies. Some thought systems, such as Buddhism and Mormonism, have no fundamental difficulty with the notion of plurality. And Christian theologians, including Protestants Paul Tillich and Krister Stendahl, Anglican C.S. Lewis, and Roman Catholic Kenneth J. Delano, predict that a leap in our knowledge of what Stendahl called “God’s world” would not be a threat, but an expanded gospel. It would enrich our sensibilities and enlarge the total

view of the world in which we live (Berendzen, 1973, p. 34; Guthke, 1990, p. 17).

Even some scientists cast ETI in a quasi-religious framework. David Viewing (1975), and D.G. Stephenson (1977) consider it possible that our solar system may be a nature reserve carefully protected by cosmic super-intelligences, for whom we are the equivalent of the wild animals in Serengeti National Park. Related views are the so-called zoo hypothesis, which we have already mentioned (Ball, 1973), and anthropologist Ashley Montagu’s notion that nuclear experiments make us so dangerous that ETI’s keep us isolated, like a potential source of infection (Berendzen, 1973, p. 21). Michael Papagiannis speculates that extraterrestrials stationed in the asteroid belt have not decided how to deal with the Earth (Papagiannis, 1978). With luck, “they” will treat our planet as a kind of cosmic nursery and prevent us from annihilating ourselves by childish pranks (Kuiper and Morris, in Goldsmith, 1980, p. 176).

These opinions are variant quasi-religious answers to a human need for a redemptive guardian, which psychologist Robert Plank locates deep in the human psyche. Erich von Däniken, author of the *Chariots of the Gods*, has been catering to it for decades. Although his assertions are considered by scientists to be dubious at best, they have been extremely profitable for him. Science fiction in the vein of *Close Encounters of the Third Kind* has fulfilled a comparable function without asserting that the claims correspond to reality.

Analogues in History

The study of three classes of analogues may be instructive to anticipation of responses by society to a confirmed detection of a signal

from an ETI. First are historic speculations about the possible existence of other worlds and public responses to these conjectures. Second and third are the responses of human societies of other centuries to intellectual communication and physical contact, respectively, with other societies.

Are we alone? Controversies about the Plurality of Worlds

As in the case of the speculative commentators on modern SETI, we can usefully divide historical opinion about the plurality of worlds into catastrophist and millenarian views (Guthke, 1990, chaps. 2-5). Catastrophists pointed to the threat that the notion of "more worlds than one" presented to presuppositions of dominant thought systems. For example, if there were other intelligent beings in space, would not the descendants of Adam and Eve lose their unique relationship with God? This anxiety lay behind the initial hostility to the concept of ETI by churches that held to the biblical account of one creation, one mankind, one fall, one Savior. Belief in other worlds and extraterrestrials was frowned upon centuries before Thomas Paine and Percy Bysshe Shelley triumphantly declared, around 1800, that it was the rock on which Christianity would founder. The flames of Hell might be preceded by those of the stake, as in Bruno's case (who, to be sure, was guilty of other theological deviations as well). Caution, as practiced by Galileo, was by far the better part of wisdom well into the seventeenth century.

Kepler put the fear of theological catastrophe in a nutshell in his *Dissertatio cum Nuncio Sidereo* (1610): "Well, then, someone may say, if there are globes in the heaven similar to our Earth, do we vie with them over who occupies the better portion of the universe?

For if their globes are nobler, we are not the noblest of rational creatures. Then how can all things be for man's sake? How can we be the masters of God's handiwork?" (Rosen, 1965, p. 43.) In Kepler's century, one took care not to undermine what he called the "revered mysteries of sacred history." Man had to be maintained at the focus of God's interest, even though, as John Donne wrote (in *The First Anniversary*):

"Freely men confess that this world's spent, When in the Planets, and the Firmament They seek so many new."

Pascal shuddered at the thought that in the sky's "infinite spaces" there were countless "realms that know nothing of us" (*Pensées*, 199, 41).

Catastrophists shifted their ground in the Age of Reason. If there are analogous intelligent beings beyond the Earth, some of them probably more advanced intellectually than we are, how can we be the measure of all things? *Homo mensura* was a widely shared self-definition of the age, which strained to leave behind the Christian concept of human nature. Man the measure was rational man, and reason the instrument and guarantee of human autonomy. The plausible conclusion that the reason of some ETI's must be superior to ours menaced man the measure. Can the Earth still be "the best of all possible worlds" if in other planet-worlds "superior beings ...shew'd a Newton as we shew an Ape," as Pope had it in the *Essay on Man* (Book II, v. 34)? Earlier Cyrano de Bergerac had turned this nightmare of reason into two satirical philosophical novels, collectively known as *Pensées; The States and Empires of the Moon and of the Sun* (Tournear and Anzieu [eds], 1662). In these celestial bodies, even the animals have a higher culture than terrestrial humans, one of whom is led on a leash like a performing

monkey in fancy dress.

The nagging fear that human rational autonomy might be just a flattering self-delusion took many forms throughout the period that produced some of the greatest triumphs of reason, culture, and science. Fontenelle's best-selling *Conversations on the Plurality of Worlds* (1686) imports a Trojan Horse through its very effort to establish that within the solar system "we" are the humanity most favored by climate and hence gifted with the greatest potential for happiness. But who can assure us that other stellar systems do not shelter far happier and luckier human races, more favored by nature (rather than by God, to whom Fontenelle pays no more than lip service)? The marquise, whose education is the object of the *Conversations*, defies anybody to dispel her cosmological depression: "Ah," she addresses her tutor, "did you but know what the fix'd stars are!" His view offers little comfort: "'Tis no great matter whether we go to them, or they come to us, we shall then be the *Americans*," that is, the weaker party (Glanvill's transl., 1688, pp. 116, 61).

The idea that Europeans amounted to no more in the cosmic order than did the unfortunate Americans on Earth recurs in Bolingbroke's "Fragments and Minutes of Essays," a work of the early 1740's, where the Enlightened philosopher suggests that the aliens' "rank in the intellectual system is even above our conceptions" (Bolingbroke, 1809, 8, pp. 173-4). The visitors from Saturn and Sirius in Voltaire's *Micromégas* (1751) laugh at the notion that Sorbonniens might have reason. The eighteenth-century poet Barthold Hinrich Brockes, terrified by his own intellectual courage, toyed with the idea that we might rank with oxen on the cosmic scale, or worse still in the view of the Hamburg burgher, with fish (Brockes, 1740, p. 322). The physiologist and poet Albrecht von Haller was shocked by

his own ability to conceive of this Earth, including his revered Switzerland, as "evil's fatherland" (Haller, 1734). In the Laputa episode in Part III of *Gulliver's Travels* (1727), Swift added the fear of physical destruction wrought by a superior civilization's space ship flying about on magnetic power.

In the nineteenth century, as Laplace's theory of planet formation, Darwin's concept of evolution, and Kirchhoff's spectral analysis made the existence of intelligent life elsewhere in the universe seem increasingly likely, Swift's vision became more plausible. H.G. Wells' story of our possible destruction by an invasion from Mars (*The War of the Worlds*, 1897) is perhaps the most familiar example of the genre. Other space novels of the second half of the century conjured up a nightmare of Martians evolved to supercivilized "degeneration" (Guthke, 1990, chap. 5., sec. 3), while Percival Lowell concluded from his observations of Mars, on which he saw evidence of a superior civilization, that "On earth, for all our pride of intellect, we have not yet progressed very far from the lowly animal state" (Lowell, 1906, p. 364). In any future encounter, we would be the Indians.

During the three centuries from Bruno to Lowell the catastrophists' litany ran to an almost continuous counterpoint of millenarian song. The God of liberal Christians of the seventeenth century, the God of reason of the eighteenth, and the "Nature" of Darwinian evolution might certainly have populated other planets with human-like beings who, in gratitude, would sing hosanna to the greater wisdom and glory of their creator. The creator need not have done the work in a literal sense; to the rationalists and evolutionists, "creator" might function as a metaphor of whatever principle they supposed to be the ultimate source of the animated cosmos. Heresy

became gospel in this perennial multi-planet chorus.

The enthusiast Bruno, who envisaged various degrees of perfection and happiness among the many planetary humanities (*On the Infinite Universe and Worlds*, 1584), sang in this chorus along with the natural philosopher Christiaan Huygens, who believed that all extraterrestrials would be exactly like us down to love of the theater (*Kosmotheros*, 1698). The English physician Robert Wittie (*Ouranoskopia*, 1681), the Protestant minister Edward Young (*Night Thoughts*, 1742-45), and the religious mystic Swedenborg (*De teluribus in mundo nostro solari*, 1758) joined in too, as did a host of poetry-writing Christian Newtonians and the anti-Christian polemicist Thomas Paine (*The Age of Reason*, 1793). Kant, Bonnet, and other believers in palingenesis (the idea that our souls, while possibly inferior to extraterrestrials now, will migrate, after death, from planet to planet, achieving greater perfection at each step) resonated to the same universe-wide hymns in praise of God as did the American scientist Alexander Winchell (*World-Life or Comparative Geology*, 1883) and the almost supernaturally indefatigable apostle of countless "humanités" in space, Camille Flammarion (*La pluralité des mondes habités*, 1862).

Whether millenarian or catastrophist, the writers we have quoted shared a belief in the existence or high probability of ETI. Of course, there have been many excellent writers in all ages who dismissed or did not bother with the possible existence of other intelligent beings in the universe. Still, as the Polish philosopher-scientist Stanislaw Lem (who imagined the possibility of many mutually unintelligible forms of "reason" in an inhabited universe) put it in his *Summa Technologiae* (1976, p. 130): "In the second half of the twentieth

century one can hardly be a complete human being without at least occasionally calling to mind that community of [extraterrestrial] rational beings, as yet unknown, to which we presumably belong."

Encounters On Earth: Transmission Of Ideas

The human experience most obviously relevant to anticipating the reaction to a SETI detection is the impact of the flow of information between terrestrial civilizations. This is especially true of non-contemporary civilizations. (Although the putative extraterrestrials may be contemporary with our civilization, because of the distances involved they are unlikely to be communicative in real time.) The historian Arnold Toynbee had a useful phrase for these interactions – "encounters between civilizations in time" – which he linked with renaissances in philosophy, religion, and art (Toynbee, 1954). One example familiar to Western intellectual history is the loss of Greek science with the fall of the Roman Empire in the fifth century A.D. The subsequent recovery and elaboration of this knowledge by the Arabs in the eighth and ninth centuries, and its entry into the Latin West in the twelfth and thirteenth centuries, provide an intriguing model for the potential impact of SETI (Grant, 1971; Lindberg, 1992). The recovery of the works of Aristotle and others led to what Haskins (1927) called "the renaissance of the twelfth century." The small army of translators employed in this old recovery of lost learning may find an analogy in the legions of scientists, cryptographers, linguists and others sure to be involved in any attempt to decipher an extraterrestrial signal. Certainly the international character of the medieval translators – Burgundio of Pisa, Adelard of Bath, Arnold of Barcelona, Philip

of Spain, Herman the German – will be replicated in the analyzers of messages from space.

The impact of an ETI detection will depend strongly on the character of the signal. A real flow of information brought about by the decipherment of an information-laden signal could prompt something like the twelfth-century renaissance. Like the medieval philosophers, we may acquire a worshipful respect for a wisdom more ancient than ours, without knowing whether it constitutes the long-sought “objective knowledge” that will propel the human species into a new age or a wisdom so superior that it will leave no scope to future human endeavor. And again, like the medieval philosophers who felt intellectually inferior to the Arabs at first encounter but grew ever more arrogant as they mastered their wisdom, we may come to challenge any ETI’s unlucky enough to undertake our education.

An example may be illustrative. In 1186 there occurred a conjunction of all the planets in Libra. That boded ill. In particular, the coming together of Jupiter and Saturn marked a change of venue of their coincidences, from one triplet of zodiacal signs to another. This vicissitude occurs every 200 years or so and always presages a calamity for someone – the French Revolution was one, later in the series. The triplicity that became the venue of conjunctions of Saturn and Jupiter in 1186 was associated by Islamic astrological doctrine with the element of air. Hence the star gazers of Toledo predicted tornadoes, sand storms, earthquakes (a global breaking of wind), and other disasters. Europeans, who had just become acquainted with the profundities of the Arabic art, took the predictions literally. They built the equivalent of bomb shelters, fasted, prayed, and wrote to Toledo for further measures. They learned that the best preparation was to become Muslims. A century later,

though still attentive to astrology, the West had found a way to moderate the fatalism of the Islamic version (“the wise man [says Saint Thomas] rules over the stars”) and had begun to develop an independent capability in exact astronomy that would soon lead them beyond their Arab mentors (Tester, 1987, pp. 148-9, 177).

The history of science contains many instructive examples of the course of assimilation of seminal ideas. The geocentric world view of the Greeks, which was based solidly on celestial phenomena, underwent several stages of elaboration, opposition, exploration of philosophical implications, and general acceptance before it was discarded in the sixteenth and seventeenth centuries. The Copernican theory, an alternative explanation of the same celestial phenomena published in 1543, also went through stages of elaboration, opposition, exploration of implications, general acceptance, and final confirmation (Kuhn, 1957). One can follow other cosmological world views through similar stages, and suggest that the putative extraterrestrial world view or “biophysical cosmology” (Dick, 1991), will follow a similar path. As we learn from the development of Copernican cosmology, the path from first presentation (1543) to general acceptance (1700) may be long, and the time to definitive confirmation (the measurement of stellar parallax in the 1830’s) even longer.

The Darwinian revolution provides a more recent example of the trajectory of an idea that bears directly on humanity’s place in nature. Many recent books analyzing the history and philosophy of Darwin’s impact might profitably be examined in relation to possible impacts of the detection of an ETI signal. There are, of course, differences in the cognitive status of a *theory* of evolution by natural selection and a *discovery* of an artificial signal from outer space. Yet the interpretation of a

signal is likely to be ambiguous and debatable. Just as Darwin's theory had distinctive impacts over the short term (Vorzimmer, 1970) and the long term (Bowler, 1989), and among scientists (Hull, 1973), theologians and other segments of the population, so probably will the discovery of extraterrestrials.

The uproar created by Darwin may not be mimicked by a SETI detection. An alternative analogy is the silent revolution ushered in by astronomer Harlow Shapley's discovery that our solar system lies at the periphery of the Milky Way Galaxy. Astronomers celebrated the discovery; the press routinely reported it; and the general population went about its business as usual despite humanity's slide from the center to the edge of the galaxy (Bok, 1974, Berendzen, 1976). Copernicus and Darwin may have inured humans to marginalization. On the other hand, the evolution of life in the universe has a more obvious connection with Darwinism than with the geography of the galaxy, and the detection of a signal from extraterrestrials might well be assimilated like Darwin's discovery rather than like Shapley's. As with other scientific theories, the impact of a successful detection by SETI will vary with different levels of society and may be absorbed over a much longer time period than we might at first imagine.

Encounters on Earth: Confrontations between Cultures

Physical contact and communication by radio are so different that analogies drawn from face-to-face encounters may not be useful in reasoning about SETI. In the terminology we introduced earlier, such analogies have more negative aspects than positive ones. Nonetheless, consequences of face-to-face encounters comprise so much human experience that it would be spendthrift to dismiss the

possibility that some lessons useful to SETI can be drawn from them. Two examples will suggest the range of lessons: a case of contact between native peoples and Western Europeans that proved beneficial for both parties and a case of conceptual misunderstanding that illustrates the difficulty that societies constructed on different principles may have in communicating with one another.

Our first example may give pause to those who think that the psychological and cultural demoralization that occurred among Native Americans and other peoples affected by European expansion should warn off would-be communicators with extraterrestrials. The proffered analogy does not take into account that the psychological impact on native peoples attributable solely to their confrontation with more technologically advanced societies cannot be separated from the wholesale degradation visited upon them by the invasion and takeover of their homelands. Most reports of "demoralized natives" have come from precisely those areas where the people have lost the most through contact with the West, namely the Americas and parts of the Pacific and Africa where imported diseases for which the people had no natural immunity ravaged populations and where Europeans conquered the survivors and settled on their lands (Crosby, 1986). Yet, until recently, psychological malaise among the survivors was often attributed to an inability to cope with the "advanced" ways of their colonial masters rather than to the violence that had been visited upon them. For example, W. H. R. Rivers (1922, pp. 84-113) thought that the Melanesian islanders of the Southwest Pacific were dying out primarily because of their "loss of interest in life" consequent upon their encounter with an all-powerful West, instead of attributing their depression and disorientation primarily to the havoc wrought by

imported epidemic diseases and colonial subjugation.

Our example of profound misunderstanding between cultures is taken from the behavior of the inhabitants of the coastal lowlands of New Guinea. There European contact produced "cargo cults," dedicated to the acquisition of *kago*, to use the Melanesian pidgin word for Western manufactured goods. Frustrated by their inability to gain much of these goods from the traders who insisted that they work many days for even a pair of shorts or a bush knife, and totally ignorant of the processes of their manufacture, Melanesian groups began developing their own solutions for gaining cargo (Lawrence, 1964). According to their cosmologies, the ancestors had provided them with their technology of stone, wood, and fiber. Therefore, the ancestors must also have designed this marvelous new technology. The Europeans had somehow interrupted the ordained line of transmission from the ancestors and were keeping most of what rightfully belonged to the Melanesians.

The Melanesians resorted to an instructive remedy to gain their *kago*. They quit working on European plantations, ceased planting their own crops, and destroyed their traditional tools and utensils. Then they performed ritual acts designed to attract ships or planes laden with cargo. They cleared landing fields and communicated with the ancestors, placing their order for cargo through dance and other means for entering the golden age of unlimited cargo.

The instructive feature of this story is the utter misunderstanding by the natives of the nature, purpose, and creation of *kago*. Why should we do any better with messages from an ETI? If Melanesians and their European colonists – creatures from the same planet and of the same biological species – could entertain such

divergent views about the derivation of technology and the acquisition of its fruits, what prospects have we for reaching ready understanding with another intelligent species from a distant star system with a biological base, evolutionary history, and cultural development entirely different from our own? Especially if, as Morrison supposes, most of the ETI's "very complex signal will contain not mainly science and mathematics but mostly what we call art and history" (Morrison, 1973, pp. 337-8). According to Morrison, we shall not be able to grasp the message they beam at us unless, in the colloquial as well as in the technical sense, we are on their wavelength.

The main message of Stanislaw Lem's *His Master's Voice* (1983) and Carl Sagan's *Contact* (1985) seems to be that contact with an ETI will not end in mutual understanding. If we credit their view, a successful outcome for SETI, the detection of the long-sought signal, might be more frustrating than exhilarating. Unable to understand or to communicate with detected ETI's, we would know that, in a profound sense, we are alone in the universe.

The experience of Asian powers like China and Japan lends little support to apocalyptic visions of the impact of contact. To be sure, both societies experienced wrenching changes from contact with the West. Yet both survived and have accommodated to the global society produced by European expansion. The Chinese had to be convinced by force of arms of the "superiority" of Western products and the "benefits" of importing them (Sahlins, 1988; and Peyrefitte, 1992). Now, not only do they continue to disappoint those who dream of exporting to their huge internal market, but they themselves are working to become major exporters in the world economy. Japan is much farther along in turning the tables on the West. After an initial flirtation with Western goods (notably firearms) and religion in the

seventeenth century, the Japanese realized the threat to their society and shut themselves off from the West until forced by Admiral Perry to open their borders in the later nineteenth century. Thereafter they set out to beat the West at its own game while staying Japanese. After their aborted attempt to become the new colonial masters of Asia, this strategy paid off brilliantly, much to the chagrin of Western leaders who now see their countries' trade deficits with Japan mounting higher and higher. What is more, the current resurgence around the world of long-suppressed nationalities and ethnic groups, as evidenced in the break-up of the Soviet Union, provides further testimony to the resilience of the human spirit even after centuries of subjugation.

The experience of the people of the remote highlands of New Guinea allows us to distinguish the psychological impact of confrontation from the degradation of exploitation. Europeans first met the inhabitants of the coast and lowlands of New Guinea in the sixteenth century and colonized them during the nineteenth century, with the usual unhappy consequences. Because of the tortuous terrain of the island's interior and the poverty of colonial budgets, however, Europeans did not contact the million or so people living isolated in high valleys scattered along New Guinea's mountainous spine until the 1930's. The stone-age highlanders quickly overcame their shock at meeting gold miners and government patrol officers carrying guns and arriving by airplanes (Connolly and Anderson, 1988). This twentieth-century encounter between an indigenous people and Europeans was much more benign than meetings of earlier centuries. Western medical knowledge and enlightened views about colonial administration spared the highlanders much of the suffering inflicted on previously contacted peoples. For example, the mass administration of sulfa drugs during World War II stopped a *Shigella*

dysentery epidemic brought by invading Japanese troops that threatened to destroy the highlanders.

By the 1970's — just forty years after initial contact — the highlanders on the eastern half of the island had gained their independence from Australia and in places were earning good money from coffee and other export crops, running their own businesses, and enjoying a wide range of consumer goods (Finney, 1973, 1986). Even outside observers who mourn the passing of the old way of life have had to admit that these highlanders, far from being disoriented and depressed by their sudden plunge into the twentieth century, have resiliently seized upon the technology and opportunities made available by their late and relatively benign contact with the outside world (Read, 1986).

If we cast ourselves in the role of the New Guinean highlanders, we might find plausible the upbeat announcements of success in SETI as described in *The Cassiopeia Affair*. Upon receiving news from radio astronomers that they had received an artificial radio signal from Cassiopeia 3579, the President of the United States in this novel addressed the general public as follows (Zerwick and Brown, 1968, p. 102):

Think of the wonderful things we can learn from our brothers of Cassiopeia. The chemistry of their life processes, the properties of their planet and their planetary system, the Galaxy as viewed from their location — all these possibilities bring tremendous excitement to our scientists.

We believe that Cassiopeian civilization is technologically more advanced than our own. Their knowledge may enable us to leapfrog over centuries of development. Cassiopeia 3579 might well provide us a sort of cosmic technology assistance program for this emerging Earth.

Above all, our Cassiopeian brothers may have found ways to solve political problems comparable to those which now threaten us on Earth. They may have found ways of managing such problems without precipitating a planet-wide holocaust such as we have feared on Earth for so long.

Recommendations

1. It is important that NASA study appropriate analogies drawn from earlier human experience, while emphasizing that they are rough guides for thinking about SETI and not precise predictors of the future.
2. Study should be concentrated on analogies based on the transmission of ideas within and between cultures in preference to analogies based on physical encounters.
3. NASA's educational programs should place SETI within the historical context of humankind's effort to comprehend its place in the universe and to understand the nature and possibility of other intelligent life.